

Unsupervised Cross-Domain Attack Traffic Classifier for Intelligent Connected Vehicle

Yongping He , Yuanqing Xia , *Fellow, IEEE*, Tijin Yan , Yufeng Zhan , Jiaru Song , and Zihang Feng 

Abstract—The heterogeneous communication ecosystem in intelligent connected vehicles (ICVs) has significantly broadened attack surfaces, making accurate attack traffic classification crucial for the security of vehicular networks. Existing deep learning (DL)-based classification methods face challenges such as reliance on labeled data and susceptibility to cross-domain data distribution discrepancies. To mitigate these challenges, a cross-domain attack traffic classifier named domain adaptation noisy label (DANL) is proposed in this article. DANL integrates uncertainty-aware sample selection and graph spectral alignment into unsupervised domain adaptation (UDA) frameworks to bridge domain gaps. Furthermore, a dual correction framework integrating noisy label learning (LNL) with pseudolabel aggregation is developed to further correct misclassifications caused by erroneous alignment. Extensive experiments on public datasets demonstrate DANL's superiority, achieving an improvement of classification accuracy up to 5.8% over baselines. Further validation in real-world tests on self-constructed ICV testbed with simulated attacks confirming its effectiveness.

Index Terms—Attack traffic classification, cyber-physical system, deep learning (DL), intelligent connected vehicle (ICV), unsupervised learning.

I. INTRODUCTION

INTELLIGENT connected vehicles (ICVs) are rapidly transforming the future of transportation by integrating advanced sensors, controllers, and communication technologies. Although this interconnected ecosystem improves safety and efficiency, it also introduces substantial cybersecurity risks. [1]. The multilayered and heterogeneous communication architecture of ICVs, including in-vehicle networks [such as controller

area network (CAN) and FlexRay], vehicle-to-everything (V2X) communication (such as DSRC and C-V2X), and cloud connectivity via transmission control protocol/internet protocol (TCP/IP), which creates a broad and diverse attack surface that is vulnerable to both conventional network attacks and cyber-physical threats [2].

Attack traffic classification technology is a pivotal tool in the network security of industrial control systems (ICS), which holds significant strategic value when applied to ICVs. As in-vehicle networks have evolved from transmitting simple control messages to encompassing sensor data and external communication signals, the dimensionality and complexity of data have significantly increased. Traditional attack classification models for single regions have struggled to meet the security requirements of complex environments. Unsupervised clustering-based methods struggle to achieve accurate classification when handling such data, as they are prone to getting trapped in local optima and sensitive to noise [3]. On the other hand, supervised deep learning (DL)-based methods have demonstrated promising classification performance in many classification tasks [4], they also face practical limitations in real-world connected vehicle environments. This is mainly due to the significant time and labor costs associated with constructing labeled datasets. Additionally, the dynamic nature of driving environments, such as variations in speed, road conditions, and weather, causes shifts in sensor data distributions, which reduces the effectiveness of static attack classification models. Furthermore, over-the-air (OTA) updates in ICVs often modify ECU software and may alter communication patterns, these changes highlight the imperative for adaptive learning frameworks capable of generalizing across diverse deployment scenarios.

Existing domain adaptation methods predominantly focus on aligning the marginal distributions of features across source and target domains, employing techniques such as adversarial adaptation or discrepancy minimization [5], [6], [7]. While effective in certain scenarios, these approaches often overlook the critical importance of sample-level uncertainty. This is particularly crucial in complex vehicle environments where data quality and reliability vary significantly. To mitigate these challenges, we propose domain adaptation noisy label (DANL), a cross-domain attack traffic classifier for ICVs based on unsupervised domain adaptation (UDA). DANL aims to achieve strong performance in attack traffic classification by integrating learning from noisy labels (LNL), all without requiring pretrained models from

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Yongping He, Tijin Yan, Yufeng Zhan, and Zihang Feng are with the School of Automation, Beijing Institute of Technology, Beijing 100081, China (e-mail: 3120215481@bit.edu.cn; yantijin@bit.edu.cn; yu-feng.zhan@bit.edu.cn; 3120185474@bit.edu.cn).

Yuanqing Xia is with Zhongyuan University of Technology, Zhengzhou 450007, China, and also with the School of Automation, Beijing Institute of Technology, Beijing 100081, China (e-mail: xia_yuanqing@bit.edu.cn).

Jiaru Song is with Jilin University, Changchun 130012, China (e-mail: songjr22@mails.jlu.edu.cn).

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other environments. This approach enhances adaptability across different network environments while maintaining robust classification accuracy. DANL constructs a graph based on feature similarities and aligns its spectral properties across domains using a graph spectral alignment loss. This facilitates a more nuanced and discriminative alignment that captures both local and global data structures, providing a more effective and flexible way to achieve domain invariance in the heterogeneous and dynamic settings characteristic of ICV environments. Furthermore, by integrating an uncertainty-aware selection mechanism that estimates the confidence of pseudolabels during adaptation, DANL can reduce the negative impact of noisy labels and enhance the overall robustness of the learning process. To mitigate inconsistencies in pseudolabels generated by different models, we frame the issue as a learning from noisy labels (LNL) problem and apply a decoupling method to reduce model discrepancies and refine the pseudolabels. Additionally, we design an aggregation mechanism to unify these pseudolabels. Through this iterative process, the pseudolabels are consolidated into hard labels and used for the next round of training. Repeating this process over several iterations leads to promising improvements in performance. The main contributions of this work are as follows.

- 1) A cross-domain attack traffic classifier DANL is introduced in this article, which enables attack traffic classification in ICVs without requiring labeled data from the specific target vehicle or environment. By integrating uncertainty-aware sample selection and alignment loss into existing UDA frameworks, DANL bridges the domain gap caused by variations in vehicle models, sensor configurations, and driving conditions, while maintaining resilience against heterogeneous data distributions common in ICV deployments.
- 2) To mitigate the inherent noise in pseudolabels caused by alignment, a dual correction framework integrating noise-resistant LNL mechanism and iterative pseudolabel refinement strategy is further introduced.
- 3) The effectiveness of DANL is evaluated through comprehensive experiments on benchmark datasets and a self-constructed ICV testbed. It achieves up to 5.8% improvement in classification accuracy compared with baselines on public datasets, while simulated attack scenario testing on the ICV testbed further confirms its robustness and practical viability for securing real-world intelligent transportation systems.

The structure of the remaining sections of this article is arranged as follows. In Section II, we detail some of the most related works. In Section III, we provide a detailed description of the main methods. In Sections IV and V, we demonstrate the results of the proposed framework by experiments. Our work is concluded in the final Section VI.

II. RELATED WORK

In this section, we give some reviews on classical works in attack traffic classification and detection, deep domain adaptation, and LNL.

A. DL-Based Attack Traffic Classification and Detection Method

DL plays a crucial role in protecting connected devices such as the internet of vehicles (IoV) and vehicular ad hoc networks (VANETs) [8], [9]. By harnessing labeled data, supervised DL-based methods exhibit remarkable efficacy, surpassing conventional methods. For example, Desta et al. [10] proposed an intrusion detection system for in-vehicle networks, which utilizes recurrence plots to generate images from in-vehicle network data and classifies by a convolutional neural network model. On the other hand, Seo et al. [11] proposed a generative adversarial network (GAN)-based unsupervised vehicle network intrusion detection model GIDS, which detects unknown attacks by learning normal data. As unsupervised methods often rely on threshold selection, leading to high false positive rates and limited to binary classification. To address this problem, Xu et al. [12] utilized a few-shot meta-learning framework to extract valuable information from a limited set of anomalous data for intrusion detection. In this article, we employ domain adaptation to extract effective information from attack traffic in other domains to assist in unsupervised attack traffic classification on the target domain. Compared with existing methods, our approach refrains from utilizing any labels from the target domain. Instead, it adeptly harnesses information from the source domain to achieve fine-grained classification of network traffic.

B. Deep Domain Adaptation

Deep domain adaptation offers a feasible solution by endeavoring to map data from disparate source and target domains with distinct distributions into a shared feature space. The primary objective is to minimize the distances of data distributions within this shared space. There are a few works concentrating on attack traffic classification and anomaly detection. Hu et al. [13] combined DSAN [14] with attention mechanism to identify malware variant traffic at an IoT edge gateway, but data used in the source and target domains are sampled from a same dataset. Wu et al. [15] proposed a multisource heterogeneous semisupervised domain adaptation method called joint semantic transfer network (JSTN) to achieve a more effective intrusion knowledge transfer. They further proposed a geometric graph alignment approach with a pseudolabel election mechanism to refine the pseudolabel of domain adaptation for intrusion detection [16]. Building models for detection and classification without any prior knowledge of target domain is first a significant challenge. In this article, we also ensemble some domain adaptation methods together to extract diverse valuable information from attack traffic in source domain to aid in classification on the target domain. Besides, we discover that when the attack traffic patterns in different domains vary significantly, relying only on domain adaptation will lead to suboptimal results. Consequently, uncertainty assessment and pseudolabels refinement are introduced to further rectify these results.

III. METHOD

In this section, the designed method for cross-domain attack traffic classification in ICVs is described in detail. We aim to

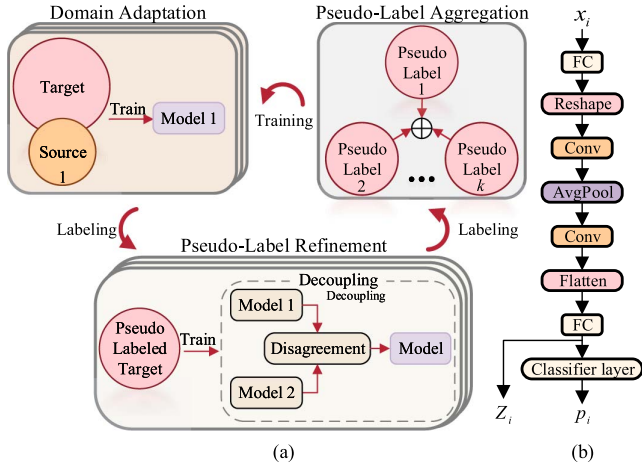


Fig. 1. Overview. (a) Workflow of DANL. Initially, different domain adaptation methods are employed to train models, which subsequently generate pseudolabels for the target data. In the subsequent phase, the decoupling method is utilized to retrain models, refining the pseudolabels of target data. Further, the pseudolabels of target data obtained by each well-trained model are aggregated in the last phase. These aggregated pseudolabeled target data are incorporated into the source data as prior information for training in next round. (b) Backbone of DANL. It takes data x_i as input and outputs the extracted feature Z_i and the probability logit p_i , which are normalized via softmax.

distill knowledge from the source domain data, where “domain” refers to different vehicles or heterogeneous communication networks in the internet of vehicles (IoV), making it applicable for identifying target domain data. The primary workflow of our designed DANL, as illustrated in Fig. 1(a), comprises three main components: domain adaptation, pseudolabel refinement, and pseudolabel aggregation. Uncertainty assessment and pseudolabel refinement are also introduced to refine pseudolabels in domain adaptation training. Furthermore, a training loss function is utilized to alleviate the alignment and class-imbalance issues. Following sections will provide detailed descriptions of each component. The backbone of the DANL can be found in Fig. 1(b).

A. UDA With Uncertainty

In the cross-domain scenario, the correlation knowledge of attack traffic is expected to be transferred from source domain to target domain, for effectively capturing the temporal dynamics and spatial structural characteristics of attack traffic. To improve computational efficiency and support parallel training, a random partitioning strategy is adopted to divide the source domain into multiple independent subdomains. Each subdomain forms a domain pair with the target domain, and UDA learning is carried out separately through parallel training. This multiperspective learning mechanism can not only accelerate the training process through a distributed computing architecture but also mine cross-domain relationships from multiple perspectives by aligning the feature distributions of different subdomains, thus enabling a more comprehensive understanding of the characteristics of the target domain.

However, in the initial training stage, due to the significant differences between the source domain and the target domain,

the training process of GAN-based models is often extremely unstable. The alignment loss $\mathcal{L}_{align}$ in DSAN is based on the proposed local maximum mean discrepancy (LMMD) criterion, which can effectively measure the difference in kernel mean embeddings between relevant subdomains of the source domain and the target domain. Therefore, a natural improvement idea is to integrate a similar metric criterion into DANN to help the model achieve stable training. Given that attack traffic contains rich temporal information and spatial structural information, we model it as a graph structure. By constructing a self-correlation graph, we can model the relationships between different traffic samples within the domain and capture the underlying data distribution characteristics.

Specifically, the backbone network f extracts features f_s and f_t from source and target domain data, respectively. An undirected weighted graph $G_s = (V_s, E_s)$ is constructed where each vertex $v_i \in V_s$ corresponds to a feature vector f_{s_i} . Edge weights $e_{i,j} \in E_s$ are determined by a distance metric $\delta(f_{s_i}, f_{s_j})$, with $\delta(\cdot)$ being a predefined metric function. The adjacency matrix of G_s is denoted as A_s . By modeling source and target domains as graphs $G_{s_1}, G_{s_2}, \dots, G_{s_k}$ and G_t , the cross-domain alignment problem is transformed into a graph matching task [17]. Given a finite graph $G = (V, E)$, the graph Laplacian Δ acting on vertex function $\phi: V \rightarrow \mathbb{R}$ and edge weighting function $\gamma: E \rightarrow \mathbb{R}$ is defined as $(\Delta_\gamma \phi)(v) = \sum_{w:d(w,v)=1} \gamma_{vw}[\phi(v) - \phi(w)]$, where $d(w, v)$ is the graph distance and γ_{vw} is the edge weight. Then for two simple nonisomorphic graphs G_s and G_t with Laplacian spectra $\Lambda_s = \{\lambda_i^s\}_{i=1}^n$ ($\lambda_1^s \geq \lambda_2^s \geq \dots \geq \lambda_n^s$) and $\Lambda_t = \{\lambda_i^t\}_{i=1}^n$ ($\lambda_1^t \geq \lambda_2^t \geq \dots \geq \lambda_n^t$), the spectral distance is $\sigma(G_s, G_t) = \|\Lambda_s - \Lambda_t\|_d$, where d denotes the order of the norm. The Laplacian matrices L_s and L_t and their eigenvalues Λ_s and Λ_t can be computed by the adjacency matrices A_s and A_t , and the graph spectral alignment loss is $\mathcal{L}_{gsa} = \|\Lambda_s - \Lambda_t\|_d$, which measures the spectral difference between domain graphs. Then for DANN, its domain classifier is $D(\cdot)$, f represents its feature extractor, the alignment loss is

$$\begin{aligned} \mathcal{L}_{align} &= \mathcal{L}_{adv} + \mathcal{L}_{gsa} \\ &= \mathbb{E}_{x_i^s \sim \tilde{\mathcal{D}}_s} \log[D(f(x_i^s))] + \mathbb{E}_{x_i^t \sim \tilde{\mathcal{D}}_t} \log[1 - D(f(x_i^t))] \\ &\quad + \|\Lambda_s - \Lambda_t\|_d \end{aligned} \quad (1)$$

where $\tilde{\mathcal{D}}_s$ and $\tilde{\mathcal{D}}_t$ denote the induced feature distributions of $\mathcal{D}_s$ and $\mathcal{D}_t$ respectively, and $\mathcal{L}_{ce}(\cdot, \cdot)$ is the cross-entropy loss function.

Furthermore, in cross-domain multiclass classification tasks, DL-based models employing cross-entropy loss are inherently prone to class bias. This bias occurs because simpler classes are learned more readily and converge faster than complex classes, even when the labels are accurate, ultimately resulting in class imbalance problems. Consequently, the following classification loss function is incorporated into our design

$$\begin{aligned} \mathcal{L}_{cls} &= - \sum_{k=1}^K q(k|x) \log p(k|x) - \sum_{k=1}^K p(k|x) \log q(k|x) \\ &\quad + \frac{\lambda}{2} \sum_{i=1}^m \|\mathbf{f}_i - \mathbf{c}_{y_i}\|_2^2 \end{aligned} \quad (2)$$

where $\mathbf{f}_i$ represents the extracted features, $\mathbf{c}_{y_i}$ refers to the center of the y_i th class associated with deep features, which can be update during training. The parameter λ balances these two components. This regularization term helps limit the distance between outliers and class centroids, which not only accelerates convergence and slightly improves performance but also distinguishes clean samples from mislabeled ones.

Furthermore, we can easily get the loss in the phase of domain adaptation as

$$\mathcal{L}_{DA} = \mathcal{L}_{cls} + \mathcal{L}_{align}. \quad (3)$$

Furthermore, existing UDA methods generate pseudolabels for the target domain. However, pseudolabel errors tend to accumulate during iterations. To mitigate noise in early iterations and avoid error propagation, we introduce an entropy-based uncertainty measure during training. This augmentation strategy enhances alignment by alleviating error accumulation in pseudolabels. The uncertainty H_i of each sample is calculated as

$$H_i = - \sum_j p_{ij} \log(p_{ij}) \quad (4)$$

where p_{ij} denotes the probability of sample i belonging to class j as predicted by the final classifier layer. The uncertainty-sorting mechanism is introduced to progressively remove data with higher uncertainty.

B. Pseudolabels Refinement by Learning From Noisy Labels

Despite enhancing pseudolabels through ensemble methods, well-trained models face challenges aligning the source and target domain distributions due to these disparities. As a result, the pseudolabels retain numerous inaccuracies. To tackle this, we employ LNL methods to transform the task of aligning data distributions into robust learning from noisy labels. It's worth noting that the noisy labels within the pseudolabel dataset of the target domain, acquired through the proposed ensemble domain adaptation method, represent a prevalent type of real-world label noise. In our studying scenario, this noise stems from the diverse knowledge acquired by each model from the source domain and the inherent disparities found within the target domain data.

After obtaining abundant data with pseudolabels by domain adaptation, a natural idea is to filter out certain data points that might be irrelevant or contaminated. We use a simple but effective decoupling method introduced in Malach et al. [18] to train models by their disagreement. We found that this decoupling method could stably improve domain adaptation results, although it gradually tends to converge. When trained on noisy labels, DL-based model first fits data with clean labels during an early learning phase. However, it then tends to memorize examples annotated with erroneous labels, consequently limiting the model's improvement. Specifically, we train two identical models h_1 and h_2 on target domain data with pseudolabel obtained from the previously well-trained model in the aforementioned domain adaptation. During the later phases of the training process, an update rule U is adopted for updating models only when

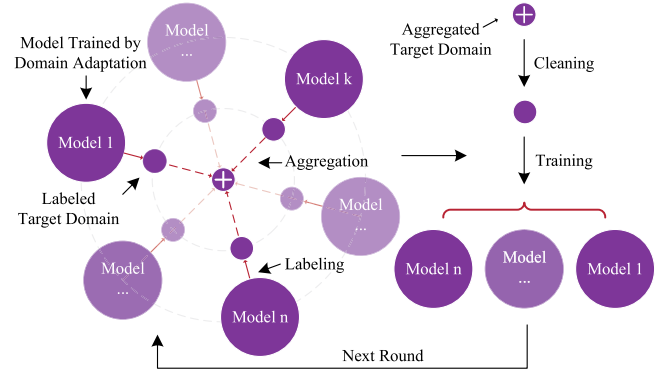


Fig. 2. Pseudolabel aggregation strategy. The pseudosoftware labels of the target domain are generated by the trained model. Redundant data are then removed through a standardized preprocessing pipeline. Conflicting annotations are resolved by majority voting. The refined labels are converted into hard labels to form a labeled dataset, which is aggregated with the original source dataset for next-round training by domain adaptation.

their judgment are in disagreement ($h_1(x_i) \neq h_2(x_i)$). For each data sample (x_i, y_i) , the update strategy can be explained as follow:

$$\begin{aligned} S &= \{(x_i, y_i) : h_1(x_i) \neq h_2(x_i)\} \\ h_1 &\leftarrow U(h_1, S) \\ h_2 &\leftarrow U(h_2, S) \end{aligned} \quad (5)$$

where S represents the dataset composed of the data that cause the different judgments between model h_1 and h_2 , and $h \leftarrow U(h, S)$ represents the model h updating with dataset S , which is usually the standard gradient descent method. For simplicity, we directly employ the backbone used in domain adaptation as the model for training.

C. Pseudolabel Aggregation Strategy

We further employ an approach to improve our model's performance in target domain through ensemble learning, which can be found in Fig. 2. After the pseudolabel refinement, each model can provide good predictions in the target domain by integrating insights from their individual source domains.

First, we directly leverage these trained models to generate pseudo soft labels (per-label logit vectors) within the target domain. Subsequently, per-label logit vectors from each model are aggregated to derive the global logit vectors. This step facilitates the assimilation of transferable knowledge from multiple source domains, enhancing classification capabilities in the target domain. Then the global logit vectors are converted into hard labels, constituting a novel pseudolabeled dataset with the corresponding features. Following this, each model is re-trained with domain adaptation by incorporating this pseudolabeled dataset alongside their respective source domain datasets. Finally, models are trained by employing the target domain data with the aggregated pseudolabels. This whole process can alleviate the negative impact of irrelevant source information, enabling the model to refine its initial judgments and achieve improved performance of classification. Despite the inclusion of diverse judgments from various models, potentially containing inaccurate pseudolabels, within the source domain data for

Algorithm 1: EM Algorithm for the Pseudo-Missing Strategy.

input : Labeled source domain data $\mathcal{D}_s$, Unlabeled target domain data $\mathcal{D}_t$; Domain adaptation method $\mathcal{M}$

output : Well-trained models $\{\theta_1, \dots, \theta_K\}$

```

1 do
2   Partition  $\mathcal{D}_s$  into  $N$  subdomains  $\{\mathcal{D}_s^1, \dots, \mathcal{D}_s^N\}$ 
   randomly with predefined seed;
   // Domain Adaptation
3   for each model  $\theta_k \in \{\theta_1, \dots, \theta_K\}$  do
4     for each subdomain  $\mathcal{D}_s^i \in \{\mathcal{D}_s^1, \dots, \mathcal{D}_s^N\}$  do
5       Train  $\theta_k$  using  $\mathcal{D}_s^i$  and  $\mathcal{D}_t$  via  $\mathcal{M}$  with
       uncertainty-sorting (Eqs. 1-4);
6       Pseudo soft labels  $\{\mathbf{y}_t^1, \dots, \mathbf{y}_t^K\}$  for  $\mathcal{D}_t$  are
       generated by the trained models;
7     end
8   end
   // Decoupling
9   for each model pair  $(\theta_i, \theta_j)$  where  $i \neq j$  do
10    Train  $\theta_i$  and  $\theta_j$  using  $\mathcal{D}_t$  with pseudo-labels
    via Eq. 5;
11    Pseudo soft labels  $\{\mathbf{y}_t^1, \dots, \mathbf{y}_t^K\}$  for  $\mathcal{D}_t$  are
    generated by the trained models;
12  end
  // Pseudo-label Aggregation
13  These source domain datasets with pseudo-labels
  are aggregated and processed to obtain global
  pseudo-labels  $\mathbf{Y}_t$ ;
14  Convert  $\mathbf{Y}_t$  to hard labels  $\hat{\mathbf{y}}_t$  to create
  pseudo-labeled dataset  $\hat{\mathcal{D}}_t$ ;
15  Update  $\mathcal{D}_s \leftarrow \mathcal{D}_s \cup \hat{\mathcal{D}}_t$  for each subdomain;
16 while training not converged;

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subsequent training iterations, it is found that models' performance will continue to improve [19], after iterating this process several times, our models have a certain ability to classify the target domain by domain adaptation.

The entire workflow consists of three phases, as shown in Algorithm 1. Initially, the labeled source domain data $\mathcal{D}_s$ is partitioned into N random subdomains $\{\mathcal{D}_s^1, \dots, \mathcal{D}_s^N\}$ using a predefined seed (Line 2). The main training loop commences with the domain adaptation phase, where each model θ_k is trained on the subdomain $\mathcal{D}_s^i$ combined with target domain data $\mathcal{D}_t$ using the specified domain adaptation method $\mathcal{M}$ with uncertainty sorting (lines 3–5). Subsequently, pseudo soft labels are generated for $\mathcal{D}_t$ by the trained model (line 6). In the decoupling phase (lines 7–9), each model pair is trained based on their disagreement regarding these pseudolabels to encourage diversity (lines 7–8). Pseudosoftware labels for $\mathcal{D}_t$ are then generated by the decoupled models (line 9). During the pseudolabel aggregation phase (lines 10–12), these pseudosoftware labels are aggregated and processed (e.g., duplicate removal) to generate global logits (line 10). These logits are converted into hard labels to create a new pseudolabeled dataset (line 11), which is then used to update each subdomain (line 12). This iterative

process continues until convergence, progressively enhancing target domain performance through collaborative model training and robust pseudolabel refinement.

IV. EXPERIMENT IN PUBLIC DATASETS

In this section, we perform comprehensive experiments on four benchmark datasets to assess the effectiveness of the proposed DANL against several baseline methods. We also examine its sensitivity to some key parameters. The ablation experiments are conducted to demonstrate the efficacy of each module within DANL as well.

A. Experimental Setup

1) *Datasets*: We utilize four benchmark network traffic datasets (KDDCup99 [20], NSL-KDD [21], CIC-IDS2017 [22], and CSE-CIC-IDS2018 [22]) to evaluate the effectiveness of the proposed DANL. These datasets are widely used in the field of network security for connected vehicles and have been proven suitable for evaluating attack traffic classification and detection performance in vehicular communication environments [23], [24]. For the aforementioned four datasets, we formulate two tasks.

The first task involves utilizing the KDDCup1999 and NSL-KDD datasets to assess the efficacy of our method across diverse data distributions. We further employ the CIC-IDS2017 and CSE-CIC-IDS2018 datasets to create a more challenging task. These datasets were collected at different times and in different environments. To ensure consistent feature dimensions, identical data preprocessing methods are applied. This allows us to assess the performance of our method when handling entirely distinct data sources with significant variations in data distribution.

2) *Baselines*: Several representative classical domain adaptation algorithms [batch nuclear-norm maximization (BNM) [25], deep adaptation network (DAN) [6], dynamic adversarial adaptation network (DAAN) [7], deep subdomain adaptation network (DSAN) [14], domain adaptive neural network (DANN) [26], graph spectral alignment framework (SPA) [27], a discriminator-free adversarial-based unsupervised domain adaptation for multilabel image classification (DDA-MLIC) (D-M, for short) [28]] are utilized to compare and confirm the effectiveness of the proposed DANL. It is worth noting that the absence of labels in the target domain renders traditional supervised intrusion detection or classification methods inapplicable, and the nuanced nature of fine-grained multiclassification problems precludes the utilization of conventional unsupervised intrusion detection techniques.

3) *Implementation Details*: To evaluate the experimental results, accuracy is adopted as a metric. Additionally, the confusion matrices is presented to provide a detailed view of the class-wise performance. In the domain adaptation phase, the model is trained with a batch size of 128 across 100 epochs, utilizing the Stochastic gradient descent (SGD) optimizer. The learning rate for the backbone is set at 1e-4, while the other layers are trained with the learning rate of 1e-3. The parameter λ in (2) is set as 1. The output dimension in Task 1 is 4, while

TABLE I
CLASSIFICATION ACCURACY RESULTS FOR TRANSFERRING BETWEEN
KDDCUP99 AND NSL-KDD

Method	KDDCup99 to NSL-KDD					NSL-KDD to KDDCup99				
	DoS	Probe	U2R	R2L	Avg.	DoS	Probe	U2R	R2L	Avg.
T-T	0.999	0.997	0.994	0.993	0.996	0.992	0.998	0.996	0.997	0.996
S-T	0.973	0.973	0.913	0.978	0.959	0.994	0.986	0.879	0.856	0.929
BNM	0.943	0.922	0.758	0.924	0.887	0.994	0.974	0.932	0.612	0.878
DAN	0.972	0.968	0.846	0.965	0.938	0.994	0.991	0.908	0.893	0.947
DAAN	0.975	0.971	0.902	0.896	0.936	0.995	0.994	0.913	0.894	0.949
DSAN	0.970	0.969	0.916	0.947	0.951	0.995	0.994	0.902	0.896	0.947
DANN	0.972	0.970	0.848	0.966	0.939	0.994	0.994	0.913	0.892	0.948
SPA	0.989	0.979	0.963	0.982	0.978	0.990	0.999	0.942	0.937	0.967
D-M	0.986	0.965	0.970	0.962	0.971	0.995	0.992	0.921	0.924	0.958
DSAN+	0.973	0.973	0.959	0.947	0.963	0.973	0.966	0.964	0.942	0.961
DANN+	0.975	0.974	0.969	0.946	0.966	1.000	0.998	0.912	0.922	0.958
DANL	0.991	0.976	0.957	0.994	0.980	0.986	0.971	0.970	0.951	0.970

Note: Bold values are in tables I-IV identify and emphasize the results obtained using DANL, the method proposed in this paper.

TABLE II
CLASSIFICATION ACCURACY RESULTS FOR TRANSFERRING FROM
CIC-IDS2017 TO CSE-CIC-IDS2018

Method	Botnet	DDoS	DoS	BF	Infiltra	WA	Avg.
T-T	0.998	0.996	0.971	0.988	0.990	0.967	0.985
S-T	0.000	0.000	0.644	0.533	0.069	0.000	0.208
BNM	0.030	0.225	0.141	0.000	0.062	0.417	0.146
DAN	0.990	0.509	0.146	0.999	0.106	0.286	0.506
DAAN	0.961	0.388	0.206	1.000	0.272	0.569	0.566
DSAN	0.980	0.054	0.623	0.999	0.115	0.285	0.509
DANN	0.961	0.001	0.220	1.000	0.271	0.568	0.504
SPA	0.970	0.689	0.754	0.930	0.563	0.693	0.767
D-M	0.959	0.460	0.707	0.990	0.303	0.591	0.668
DSAN+	0.961	0.037	0.690	0.999	0.115	0.286	0.515
DANN+	0.961	0.388	0.206	0.999	0.272	0.569	0.566
DANL	0.972	0.999	0.761	0.814	0.540	0.865	0.825

TABLE III
CLASSIFICATION ACCURACY RESULTS FOR TRANSFERRING FROM
CSE-CIC-IDS2018 TO CIC-IDS2017

Method	Botnet	DDoS	DoS	BF	Infiltra	WA	Avg.
T-T	0.995	0.987	0.999	0.999	0.693	0.991	0.944
S-T	0.000	0.180	0.260	0.341	0.590	0.006	0.230
BNM	0.604	0.462	0.114	0.813	0.022	0.041	0.343
DAN	0.603	0.416	0.530	0.812	0.524	0.041	0.488
DAAN	0.600	0.002	0.551	0.823	0.566	0.041	0.431
DSAN	0.940	0.508	0.846	0.814	0.471	0.049	0.605
DANN	0.811	0.507	0.600	0.814	0.453	0.095	0.547
SPA	0.969	0.787	0.892	0.821	0.562	0.082	0.686
D-M	0.941	0.613	0.747	0.816	0.593	0.052	0.627
DSAN+	0.937	0.767	0.893	0.816	0.573	0.041	0.671
DANN+	0.806	0.502	0.660	0.823	0.566	0.001	0.560
DANL	0.973	0.817	0.883	0.813	0.860	0.045	0.732

it is set to 6 in task 2. Each training iteration involves selecting the top 10% of samples based on their uncertainty for domain adaptation.

B. Effectiveness Evaluation

In Tables I-III, we demonstrate the effectiveness of DANL in attack traffic classification across four network traffic datasets, confusion matrices of these target datasets are also visualized

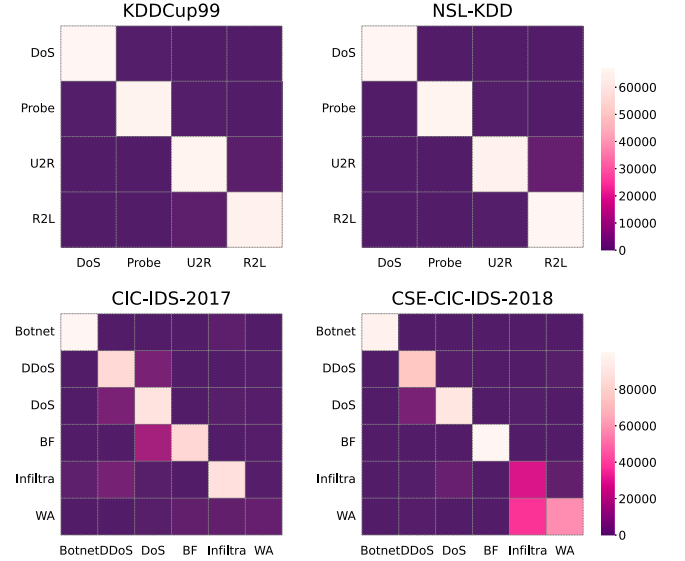


Fig. 3. Confusion matrices of the target datasets. Different colors denote diverse sample counts, where lighter shades correspond to higher counts. For the transfer tasks on KDDCup99 and NSL-KDD datasets, the diagonal values typically range from 0 to approximately 60 000. For the transfer tasks on CIC-IDS2017 and CEC-CIC-IDS2018 datasets, the diagonal values range from 0 to approximately 80 000.

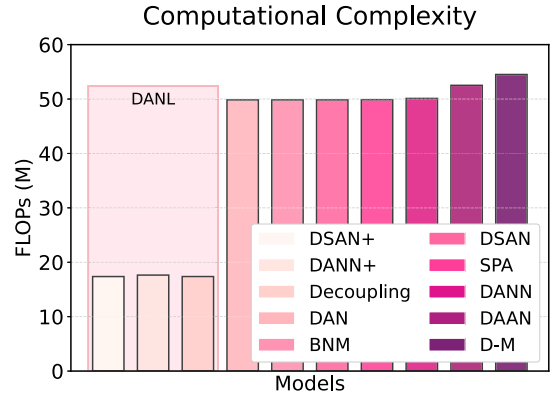


Fig. 4. Computation complexity analysis. DANL mainly consists of three modules, and the total FLOPs are represented by the pink rectangle.

in Fig. 3. We compare it with established benchmark algorithms, presenting both the overall accuracy and class-specific accuracies in these Tables. “T-T” signifies that training and testing models both on the same target domain, which is also the common practice of supervised methods. These methods rely heavily on a large amount of labeled data, which is usually impossible to achieve in practical scenarios. Notably, this “T-T” displays effective recognition of various attack traffic behaviors, achieving commendable results and representing our method’s ultimate aspiration. The “S-T” illustrates the results of training the model directly on the source domain and testing on the target domain. Although the model exhibits acceptable performance on task 1, its efficacy significantly diminishes on the more challenging task 2. Specifically, it suggests that applying knowledge solely acquired from CIC-IDS2017 directly to the CSE-CIC-IDS2018 dataset fails to identify Botnet and

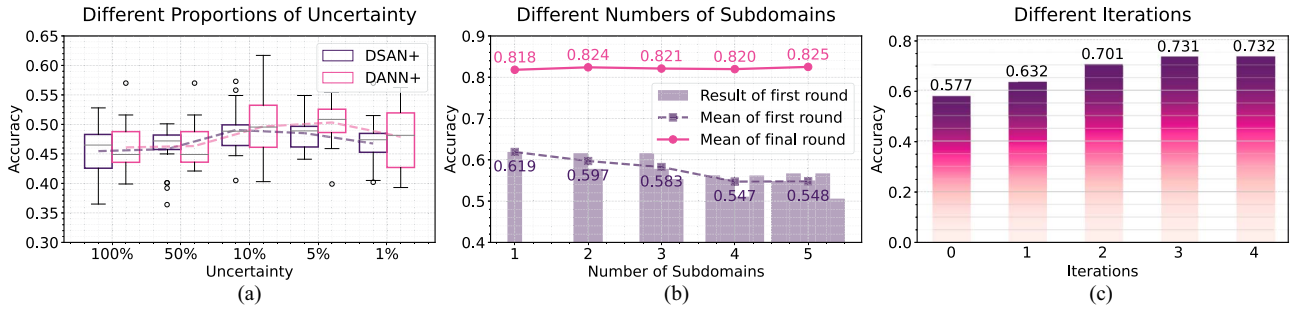


Fig. 5. Accuracy result of model robustness study. (a) Accuracy result of model robustness study with uncertainty. The boxplots depict the impact of varying uncertainty levels on the performance of two domain adaptation methods. Numerical values represent the means, while gray lines indicate medians. The trends in maximum and minimum values are illustrated by dashed connections, reflecting their fluctuations. (b) Accuracy result of model robustness study with different numbers of subdomain. (c) Accuracy result of model robustness study with iterations. The performance of models improves with increasing iterations, eventually tend to stabilize.

DDoS attacks effectively. The results of comparative algorithms are also presented in these tables, most of which surpass the “S-T”, which signifies the rationality and applicability of domain adaptation in addressing cross-domain attack traffic classification problems. Additionally, “DSAN+” and “DANN+” denote the classification results achieved by replacing the original classification loss in these two domain adaptation methods with our proposed loss function. It is evident across all tasks that these modifications yield performance improvements over the original methods, underscoring the efficacy of our designed loss function. The last row in each table showcases the results of our proposed DANL, it surpasses all comparative algorithms, thereby substantiating the effectiveness of our approach.

Furthermore, we have also provided the computational complexity of DANL compared with the baselines, as shown in Fig. 4. The designed lightweight backbone enables DSAN+ and DANN+ to achieve lower computational complexity. The decoupling module’s dual-model training roughly doubles FLOPs to 8.688M, leading to a total of approximately 52.393M FLOPs for the DANL framework. While overall FLOPs are comparable to baseline models, our modular decoupling and collaborative optimization ensure component independence and reduce individual module costs. This design balances controllable complexity with full functionality, ideal for resource-constrained domain adaptation tasks such as edge deployment and real-time processing.

C. Sensitive Analysis

To further assess the stability and robustness of our proposed model, we conduct sensitivity experiments on some critical parameters. First, we evaluate the strategy of selecting training sample based on uncertainty during the domain adaptation phase, and observe its impact on the model’s performance by adjusting the proportion of training data. As depicted in Fig. 5(a), using 100% of the target data increases training time and may reduce model performance due to the accumulation of incorrect pseudolabels. After gradually filtering out samples with lower confidence, the model performance shows an upward trend. However, when an extremely small proportion of data is used, the model performance also degrades, indicating that the model cannot fully learn effective knowledge from

the target domain due to insufficient training samples. There appears an inflection point in model performance as the proportion gradually decreases. It can be observed that for DSAN, this turning point is around 10%, while for DANN, it lies between 5% and 10%, with little difference in model performance between these two proportions.

We further investigate how partitioning the source domain affects model performance in domain adaptation for the task of transferring from CIC-IDS2017 to CSE-CIC-IDS2018, which can be viewed as leveraging limited attack traffic knowledge from multiple vehicles to assist classification in the target vehicle [Fig. 5(b)]. The results reveal a gradual performance decline as the number of subdomains increases (with correspondingly fewer training samples per subdomain), though the model eventually stabilizes at an acceptable accuracy level. While this partitioning approach incurs a modest performance reduction in the initial phase, it offers significant practical advantages by enabling efficient deployment across multiple lightweight devices. This trade-off proves particularly valuable for real-world IoV applications, where the model’s ability to run concurrently on more resource-constrained edge devices (e.g., vehicle ECUs or roadside units) outweighs the marginal accuracy decrease, while maintaining sufficient classification capability for most attack scenarios. Additional experiments analyzing the impact of different random seeds on domain partitioning can be found in Fig. 6. As can be seen from the training results of the first round, compared with DSAN+, the accuracy fluctuation of DANN+ across various subdomains is slightly larger; however, the average accuracy of both models remains relatively stable under different seed settings.

Subsequently, we delve deeper into the impact of iteration rounds on the performance using the task from CIC-IDS2018 to CSE-CIC-IDS2017 as an example. As illustrated in Fig. 5(c), the results indicate that relying solely on source domain information obtained from some single domain adaptation method is insufficient for handling the complexity of extensive high-dimensional attack traffic. This inadequacy becomes evident in effective classification of attack traffic in the target domain. In addition, it can be seen from the figure that implementing only one iteration strategy can significantly improve the model’s ability. Moreover, as the round of iteration increases, the model consistently improves in classification performance. Notably,

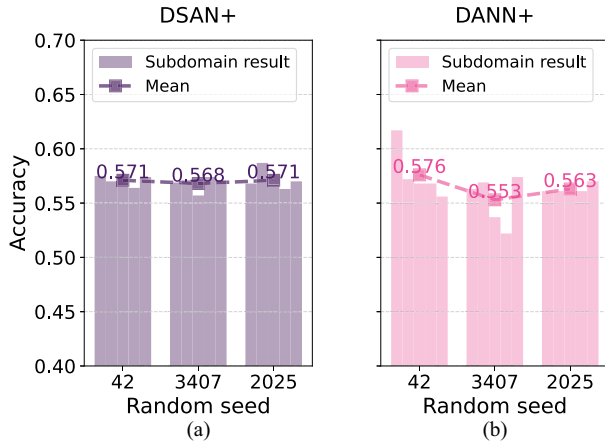


Fig. 6. Accuracy result of (a) DSAN+ and (b) DANN+ with random seeds. The bar chart represents results on each subdomain under the setting of splitting the source domain into five parts, and the line chart denotes the average value.

the performance stabilizes after approximately 3–4 rounds of iteration, validating the robustness of our iterative approach. These findings underscore the necessity and effectiveness of iterative strategies in bolstering classification capabilities and effectively classifying attack traffic during the knowledge transfer across domains.

D. Ablation Study

We also conduct ablation experiments to validate the effectiveness of the components in the proposed DANL, as illustrated in Table IV, where “1st-D” means the first round of decoupling. The original DSAN and DANN employ the standard cross-entropy loss for classification assistance. Subsequently, integrating our designed classification loss enhances the results. After the first iteration with the decoupling method, the model achieves a result of 0.568. This iteration enables the model to learn the discrimination of “DDoS” and “Infiltra” (infiltration) attack traffic from DSAN+ and DANN+. Upon involving the target domain dataset with aggregated pseudolabels in domain adaptation training, the model’s performance further improves. After three iterations, the model reaches a final result of 0.825, showcasing robust judgments across various classes. Notably, our approach contributes to providing further granularity in unsupervised methods, enabling the identification of diverse attack traffic.

We then conduct ablation experiments to compare different distance metrics in the alignment loss under the task from CIC-IDS2017 to CSE-CIC-IDS2018. The results are summarized in Table V. The results indicate that using Euclidean distance consistently outperforms other metrics for both DANN+ and DSAN+. This superiority is likely attributed to its sensitivity to magnitude differences in feature distributions, which aligns well with our Laplacian spectral framework. Based on these findings, we have adopted Euclidean distance as the default choice for empirical evaluations, given its demonstrated effectiveness.

TABLE IV
ABLATION EXPERIMENTS FOR TRANSFERRING FROM CIC-IDS2017 TO CEC-CIC-IDS2018

Method	Botnet	DDoS	DoS	BF	Infiltra	WA	Avg.
DSAN	0.772	0.040	0.670	0.999	0.117	0.145	0.457
DANN	0.767	0.095	0.516	0.999	0.062	0.137	0.429
DSAN+	0.961	0.037	0.690	0.999	0.115	0.286	0.515
DANN+	0.961	0.388	0.206	0.999	0.272	0.569	0.566
1st-D	0.961	0.747	0.231	0.999	0.391	0.557	0.568
2nd-DSAN+	0.961	0.757	0.889	0.999	0.172	0.569	0.725
2nd-DANN+	0.961	0.734	0.404	0.999	0.211	0.552	0.644
2nd-D	0.960	0.739	0.883	0.999	0.286	0.583	0.742
3rd-DSAN+	0.961	0.757	0.900	0.999	0.285	0.582	0.747
3rd-DANN+	0.961	0.759	0.901	0.999	0.298	0.583	0.750
3rd-D (DANL)	0.972	0.999	0.761	0.814	0.540	0.865	0.825

TABLE V
ABLATION STUDY ON THE EFFECTIVENESS OF DISTANCE METRICS FOR THE ALIGNMENT LOSS

Model	Distance Metric		
	Cosine	Euclidean	Gaussian
DANN+	0.520	0.566	0.534
DSAN+	0.479	0.515	0.512

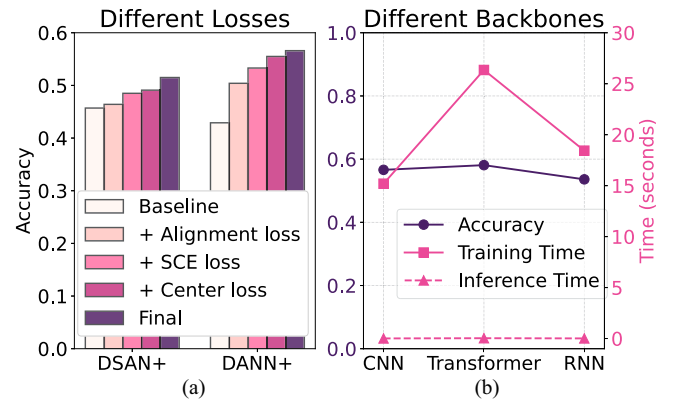


Fig. 7. Ablation study on: (a) loss; and (b) backbone component effectiveness.

To further investigate the impact of each loss component on the results we conduct study under the task of transferring from CIC-IDS2017 to CSE-CIC-IDS2018, as shown in Fig. 7(a), where “baseline” refers to the original DSAN and DANN models. It can be observed that the alignment loss appropriately improves the model performance during the first round of training, while the addition of more robust classification losses helps better distinguish decision boundaries and enhances classification accuracy. Furthermore, we conduct experiments to show the results of replacing the designed CNN backbone in DANL with RNN and Transformer, as presented in Fig. 7(b). It can be observed that DANL, with its designed lightweight CNN-based backbone, achieves detection results comparable to those of the Transformer while maintaining relatively low training and inference time per epoch, indicating its potential for application in edge or lightweight devices in the ICVs.

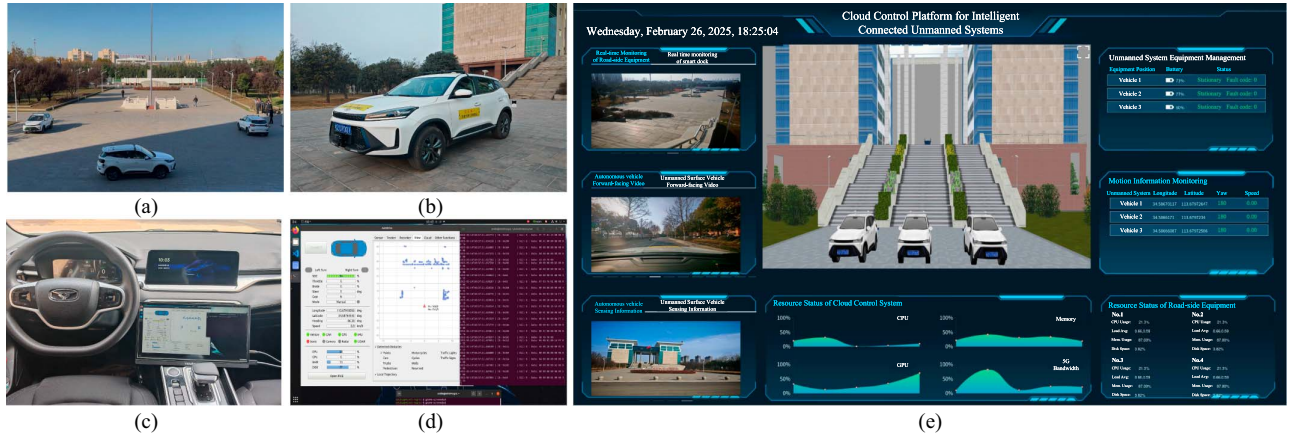


Fig. 8. Cloud control platform for intelligent connected unmanned systems. (a) Outdoor operating scene of autonomous vehicles. (b) Close-up of autonomous vehicle exterior. (c) Interior view of unmanned vehicle cockpit. (d) Data monitoring interface of unmanned vehicle system. (e) Interface of cloud control platform for intelligent connected unmanned systems.

V. EXPERIMENT IN REAL-WORLD SCENARIO

In this section, we verify the effectiveness of the proposed method by collecting data from real vehicles on a self-construct intelligent connected unmanned platform.

A. Experimental Setup

The experimental platform is shown in Fig. 8. The vehicle-road-cloud integrated control system part of this platform consists of three unmanned vehicles, a cloud data center, and supporting monitoring equipment. Fig. 8(a)–(b) depicts the experimental scene and the full-view exterior of the autonomous vehicle (Kaiyi Showjet Pro EV), while Fig. 8(c)–(d) shows the interior of the cockpit, including the steering wheel, dashboard, and related display devices. Each autonomous vehicle is equipped with NVIDIA Orin. The autonomous driving system operates on the Ubuntu 20.04 operating system, with robot operating system2 (ROS2) framework, and the algorithms are developed using Python 3. The system adopts a distributed and modular design, covering core functional modules such as perception, planning, control, coordination, status monitoring, positioning, and map building. These modules are both independent and work in coordination, enabling the efficient completion of autonomous driving tasks in complex scenarios. Fig. 8(e) displays the real-time monitoring interface of the cloud control platform. Multisource data (environmental perception, device status, and vehicle trajectories) are collected by the devices deployed on roadside sensors and onboard terminals. These data are transmitted via 5G/4G networks to the platform. A digital twin of physical road scenarios and vehicle operations is dynamically updated through 3-D modeling and simulation techniques. Structured information (e.g., vehicle speed and fault alerts) and raw data are stored in Alibaba Cloud databases for efficient storage, querying, and historical traceability.

The CAN protocol is a serial communication protocol designed for real-time communication, with wide applications in automotive electronics, industrial automation, and other fields.

Considerable research has been conducted on intrusion detection and attack classification for the CAN, as referenced [29], [30], etc. We collect real-time CAN data from three autonomous vehicles during their operation at different times to obtain datasets for various scenarios. Similar to existing works [11], [29], [31], three types of attacks are considered: DoS attack, fuzzy attack, and injection attack.

B. Effectiveness Evaluation

The effectiveness of the proposed DANL is evaluated using the collected CAN dataset, with the corresponding experimental results presented in Fig. 9. The figure illustrates the attack traffic classification accuracy and F1 scores under various transfer learning settings. For instance, following the experimental setup described in Section IV, the label “From-Car1-to-Car2” on the horizontal axis represents the scenario where attack detection is performed on the second vehicle using knowledge extracted from the first vehicle. The classification accuracy and F1 score results suggest that the proposed DANL exhibits competitive attack classification performance across most transfer settings, indicating its potential as a viable solution for attack detection in heterogeneous vehicular systems. Additionally, confusion matrices for target datasets across three autonomous vehicles are presented in Fig. 10. High diagonal values of these matrices indicate accurate classification of specific attack types, highlighting the model’s reliability in threat detection. Low off-diagonal entries further suggest minimal misclassification, underscoring its robustness and capacity to distinguish traffic patterns across heterogeneous vehicular systems.

Furthermore, to evaluate the attack traffic classification performance of DANL in cross-vehicle scenarios, experiments are conducted using the public dataset OTIDS [31] and our dataset. Taking the first car (car 1) as an example, the results are shown in Fig. 11(a). The results of DNAL, first-round DSAN+, DANN+, and direct training and testing (“S-T”) are provided. The first-round results of DSAN+ and DANN+ reflect initial improvements through migration over direct training and testing. DSAN+ shows limited attack classification performance

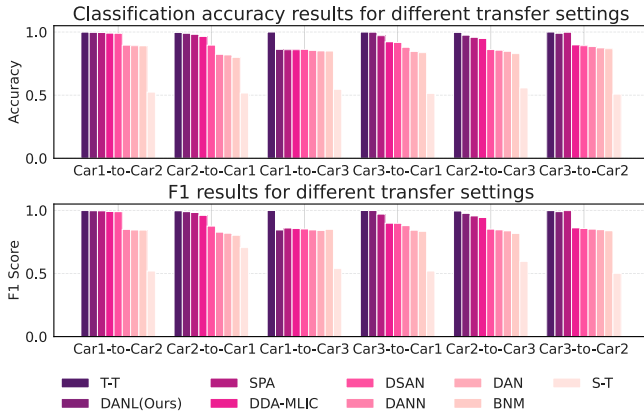


Fig. 9. Comparison of accuracy and F1 results with different transfer settings in CAN datasets.

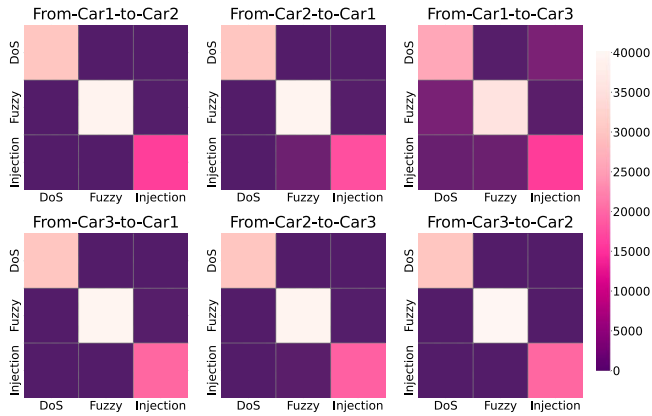


Fig. 10. Confusion matrices of the target datasets across three different autonomous vehicles. For the transfer tasks between the three vehicles, the diagonal values in the confusion matrices are generally in the range of 0 to approximately 40 000.

when transferred from the OTIDS dataset to car 1. After incorporating the knowledge from DANN+ and applying iterative optimization, DNAL achieves significantly improved classification accuracy.

We further investigate the impact of cross-network data on the performance of attack traffic classification. Specifically, we transfer knowledge between CAN data and external sensor network data to simulate real-world scenarios. As illustrated in Fig. 11(b), we extract historical sensor network data from the Alibaba cloud database, which captures environmental conditions and traffic patterns. We then conduct transfer learning experiments with the CAN data from car 1. The results show that direct training on one network and testing on the other yield accuracies of only 0.492 and 0.373, highlighting the challenge of cross-network generalization, while DANL maintains stable classification accuracy in both scenarios. This demonstrates DANL's superiority in handling heterogeneous data sources, suggesting its potential as a viable option for securing inter-connected vehicle systems in practical deployments.

Finally, we consider a new type of attack: a message injection attack where we inject tampered data to switch the vehicle's

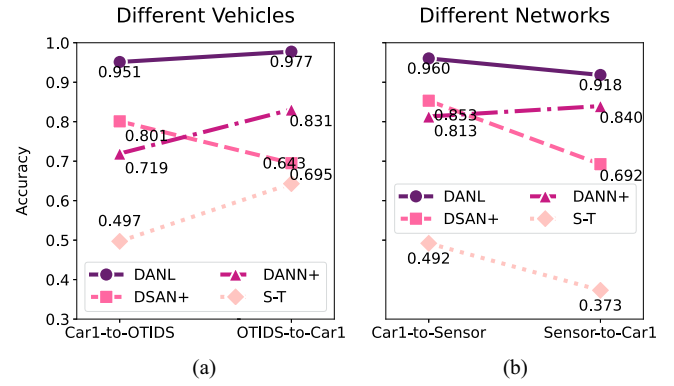


Fig. 11. Classification performance results under: (a) different vehicles; and (b) different networks.

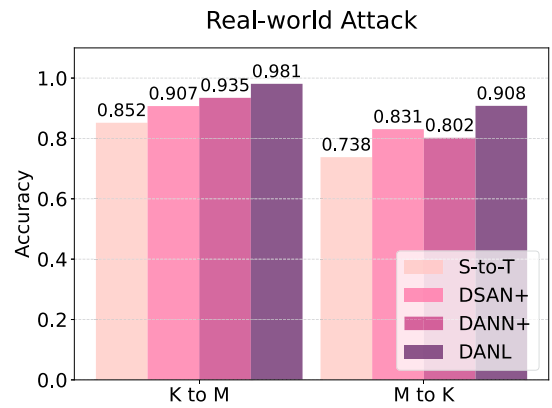


Fig. 12. Classification performance results under the designed attack. The notation 'K to M' represents the transfer learning scenario from the Kayyi Showjet Pro EV (denoted as K) to the Wuling Hongguang Mini EV (denoted as M), with the reverse transfer indicated accordingly.

high and low beam headlights. In real-world nighttime highway conditions, sudden frequent switching of high and low beams can indeed distract and impair the vision of oncoming drivers, potentially leading to safety hazards. Since this attack primarily affects the protocol related to the vehicle's lighting system, it poses relatively low risk during experiments. Specifically, we implemented this by alternately injecting CAN messages that switch the headlights during normal driving conditions and collected corresponding data. To simulate cross-domain transfer scenarios, we included an additional vehicle of a different model, the Wuling Hongguang MINIEV. The collected dataset comprises three types of attack data: 1) DoS attack data; 2) fuzzy attack data; and 3) message injection attack targeting the lighting system. The results of experiments in transferring between Kayyi (K) and MINI (M) can be found in Fig. 12. It can be observed that, even though there are differences in the CAN IDs and message formats associated with headlight control among these distinct vehicle models, the cross-domain feature alignment strategy of DANL sustained robust attack classification performance in both scenarios. This serves to validate the robustness of our proposed method with respect to protocol variations across heterogeneous vehicles.

VI. CONCLUSION

In this article, we proposed a cross-domain attack traffic classifier for ICV named DANL, which leveraged attack traffic information from other domains to aid the unlabeled target domain in fine-grained attack traffic classification. It amalgamated various domain adaptation methods to train models in the source domain, acquiring pseudolabels for target domains and subsequently refining these labels via learning from noisy labels. By integrating a pseudolabel aggregation strategy, a straightforward iterative process showcased promising performance. Experimental results showcased that the proposed DANL outperformed existing methods by up to 5.8% in performance across classical network traffic datasets. Moreover, the effectiveness of DNAL has been verified by real data experiments on the self-constructed ICV platform.

It can be found that DANL can be integrated with existing unsupervised anomaly detectors to perform fine-grained network intrusion detection. After intercepting the data, most of the easily identifiable normal traffic is filtered by traditional unsupervised methods with set thresholds, and then DANL is used to further classify the attack traffic data in the target domain by learning the knowledge from other domains. In the future, we will explore more appropriate data augmentation methods for attack traffic classification, specifically addressing class imbalance. Furthermore, we will investigate the incorporation of multimodal data sources to enrich feature representations and enhance detection capabilities. We will also consider more practical and theoretically guaranteed adversarial attack models, continuously improving our classification and detection methods from an adversarial perspective.

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Yongping He is currently working toward the Ph.D. degree in control science and engineering from the School of Automation, Beijing Institute of Technology (BIT), Beijing, China.

His research interests include Internet of Things, time series anomaly detection, and generative models.



Yuanqing Xia (Fellow, IEEE) received the Ph.D. degree in control theory and control engineering from Beijing University of Aeronautics and Astronautics, Beijing, China, in 2001.

From 2002 to 2003, he was a Postdoctoral Research Associate with the Institute of Systems Science, Academy of Mathematics and System Sciences, Chinese Academy of Sciences, Beijing. From 2003 to 2004, he was with the National University of Singapore, Singapore, as a Research Fellow, where he worked on variable structure control. From 2004 to 2006, he was with the University of Glamorgan, Pontypridd, U.K., as a Research Fellow. From 2007 to 2008, he was a Guest Professor with Innsbruck Medical University, Innsbruck, Austria. Since 2004, he has been with the Department of Automatic Control, Beijing Institute of Technology, Beijing, first as an Associate Professor, then, since 2008, as a Professor. In 2023, he was also hired as the President of Zhongyuan University of Technology, Zhengzhou, China. He is currently both a Professor with Beijing Institute of Technology and the President of Zhongyuan University of Technology. His research interests include cloud control systems, networked control systems, robust control and signal processing, active disturbance rejection control, unmanned system control, and flight control.



Tijin Yan received the B.S. degree in control science and engineering in 2019 from Beijing Institute of Technology (BIT), Beijing, China, where he is currently working toward the Ph.D. degree in control science and engineering.

His research interests include graph representation learning, generative models, time series modeling, time series anomaly detection, and dynamic systems.



Yufeng Zhan received the Ph.D. degree in control theory and control engineering from Beijing Institute of Technology (BIT), Beijing, China, in 2018.

He is currently an Associate Professor with the School of Automation, BIT. Prior to joining BIT, he was a Postdoctoral Fellow with the Department of Computing, The Hong Kong Polytechnic University, Hong Kong. His research interests include Internet of Things, cloud/edge computing, and machine learning systems.



Jiaru Song was born in Shanxi, China, in 2000. He received the M.S. degree in computer science from Jilin University, Jilin, China, in 2025.

He is currently a Research Assistant at The Hong Kong Polytechnic University, Shenzhen Research Institute. His research interests include intelligent connected vehicle and intrusion detection.



Zihang Feng was born in Shanxi, China, in 1997. He received the B.S. degree in automation in 2018 from Beijing Institute of Technology, Beijing, China, where he is currently working toward the Ph.D. degree in control science and engineering with the School of Automation.

His research interests include visual tracking and multimodality fusion.